

Certifying the refined Schur at $L = 0.3$:

ε lifts, the Fourier route dies,
the tail integral remains

PRIME.RDAGGER.SCHUR_CERTIFICATE.01 — round 481

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August 31, 2026

Abstract

Round 480 reduced $\lambda_*(0.3) \geq c$ to positivity of the even 3×3 matrix Schur $S = B - CC^\top / c_{\text{tail}}$ on $\text{span}\{\psi_0, \psi_2, \psi_4\}$ (numeric $\lambda_{\min}(S) \approx 5.69 \cdot 10^{-3}$) and an easier odd partner. This round tries to enclose S . Route (b) with the integer Fourier ONB is killed: $Q(u_0, u_n) \sim 1/n$, and the infinite tail of C eats the midpoint Schur. The identity $\|Au\|_{L^2(\mathbb{R})}^2 = (2\pi)^{-1} \int \sigma_A^2 |\hat{u}|^2$ is a false formula for $\|Au\|_{L^2[-L, L]}$ and overestimates CC^* by an order of magnitude. Even Legendre polynomials have an explicit spherical Bessel Fourier transform and a tiny ε , but the same completeness leak as M grows. The working basis is the Nyström–Slepian interpolant. The kernel is entire: Gauss–Legendre $n = 48$ interpolation remainder is $< 10^{-80}$, discrete residuals are 10^{-16} , so $\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}$ and $c_{\text{tail}} \geq 0.2408$ lift from the discrete matrix (PROVED). The remaining integrals are $\int_0^T \sigma \hat{\psi}_i \hat{\psi}_j$ (oscillatory, working window $T \lesssim 200$ at $n = 64$) and the $[T, \infty)$ integration-by-parts tail. The crude envelope $|\hat{\psi}| \leq 2|\psi(L)|/t$ produces an A -tail ~ 0.76 on mode 4 at $T = 150$, larger than the Schur margin, so it is not a pad. True windows through $T = 320$ stay PD ($\lambda_{\min}(S) \approx 7.1 \cdot 10^{-3}$) but are not an interval hull. Verdict REDUCED(interval-FT+IBP-tail). Fourier–HS is not reused. Yoshida–Bombieri zone. No RH claim.

Status. REDUCED(interval-FT+IBP-tail). No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq 0$. ε and c_{tail} lift; $S \succ 0$ does not. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $P \equiv 0$. Plancherel $\hat{h}(t) = \int h(x) e^{itx} dx$, $\|h\|_2^2 = (2\pi)^{-1} \int |\hat{h}|^2$. The r479/r480 contract is in force: $t_* = 8$, c_{tail} , κ_{low} , and the named operator-norm bound is already false.

2 Basis choice

The tail floor $A \succeq \sigma_A(t_*)I + (\sigma_A(0) - \sigma_A(t_*))K_{t_*}$ is operator-wide. On $\text{ran}(1 - P)$ it gives $c_{\text{tail}} = \sigma(t_*)(1 - \varepsilon) + \sigma(0)\varepsilon$ with $\varepsilon = \|(1 - P)K(1 - P)\|$, a single Rayleigh number for whatever 3-dimensional P one chooses.

Fourier ONB (killed). $V = \text{span}\{1, \cos(\pi x/L), \dots, \cos(4\pi x/L)\}$ has $\varepsilon \approx 1.3 \cdot 10^{-4}$ and a certified 5×5 Gram (r478, Higham $\mu \geq 4.06 \cdot 10^{-3}$). The off-block $Q(u_0, u_n)$ decays like $1/n$ (0.321, 0.161, 0.111, 0.078, 0.066 for $n = 1..5$). At truncation $M = 10$ one has $\lambda_{\min}(S) \approx 1.3 \cdot 10^{-4}$; the leftover Hilbert–Schmidt mass for $n \geq 10$ is ~ 0.08 and kills the Schur. This is the r478 obstruction in refined clothing.

σ^2 -Plancherel (killed). $\|M_\sigma u\|_{L^2(\mathbb{R})}^2 = (2\pi)^{-1} \int \sigma^2 |\hat{u}|^2$ majorizes $\|Au\|_{L^2[-L, L]}^2$ but is not equal to it ($M_\sigma u$ is not time-limited). Used as an exact CC^* it produces $\lambda_{\min}(S) \approx -10$.

Even Legendre (not closed). P_0, P_2, P_4 have Fourier transforms $2Li^n j_n(tL)$ (spherical Bessel). $\varepsilon \approx 2 \cdot 10^{-6}$. As the complementary degree M grows, $\|C\|$ creeps ($0.31 \rightarrow 0.55$ from $M = 6$ to $M = 20$) and $\lambda_{\min}(S)$ hovers near $4 \cdot 10^{-3}$ without a certified infinite- C bound.

Nyström–Slepian interpolant (working). The first three even prolates are the unique small space that is almost invariant under K_8 , so C lives in one or two neighbouring modes (ψ_6, ψ_8) and the M -truncation of r480 is honest. The interpolant through the Gauss nodes, extended by $\tilde{\psi}(x) = \lambda^{-1} \sum_j w_j K(x, x_j) \psi(x_j)$, is an explicit combination of sines.

3 What is proved: the lift of ε

Lemma 1 (Kernel interpolation). *The sinc kernel $K(x, y) = \sin(t_*(x - y))/(\pi(x - y))$ is entire in each variable. The Gauss–Legendre remainder prefactor on $[-L, L]$ at $n = 48$ is $7.59 \cdot 10^{-230}$. Times t_*^{96} this is $< 10^{-80}$.*

Lemma 2 (Discrete residuals). *The weighted Nyström eigenproblem at $n = 48$ has node-wise L^2 residuals $< 10^{-15}$ on modes 0..6. Weyl plus the interpolation remainder therefore lifts*

$$\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}, \quad c_{\text{tail}} \geq 0.2408.$$

Endpoint jets $\psi(\pm L)$ of modes 0..4 agree at $n = 48$ versus $n = 64$ to 10^{-12} .

4 What is not proved: the tail integral

Double integration-by-parts gives

$$\hat{\psi}(t) = \frac{2\psi(L) \sin(tL)}{t} - \frac{2\psi'(L) \cos(tL)}{t^2} + O(t^{-3}),$$

because $\psi(\pm L) \neq 0$. The envelope $|\hat{\psi}| \leq 2|\psi(L)|/t$ produces

$$\int_{150}^{\infty} \frac{(1 + \log t) |\hat{\psi}_4|^2}{\pi} dt \sim 0.76,$$

larger than $\lambda_{\min}(S)/3$. It cannot be used as a pad.

True increments at $n = 96$ (not a hull): $A[80, 160]$ adds 0.163 to A_{44} and 0.187 to A_{46} ; windows through $T = 320$ keep $\lambda_{\min}(S) \approx 7.1 \cdot 10^{-3}$. Gauss-in- x dies past $T \sim 200$ at $n = 64$ (the ratio $|\hat{\psi}|/(2|\psi(L)|/t)$ jumps from 0.9 to 5).

The remaining object is the oscillatory integral $\int_T^\infty \sigma_A(t) \hat{\psi}_i(t) \hat{\psi}_j(t) dt$ with a Filon or cosine-integral remainder small enough to leave a third of the margin. Four even block modes instead of three does not remove it ($\lambda_{\min}(S)$ stays $\approx 6.8 \cdot 10^{-3}$ at $T = 240$).

5 Error budget

source	bound	vs margin $5.7 \cdot 10^{-3}$	status
Nyström ε lift	10^{-80}	$10^{-5}\%$	PROVED
discrete residual	10^{-15}	$10^{-10}\%$	PROVED
Weyl on 48×48	10^{-14}	$10^{-9}\%$	PROVED
c_{tail} from ε	0.2408	—	PROVED
$[0, T]$ FT at $T \leq 150$	float only	unknown	OPEN
IBP envelope pad	0.76	13000%	KILL as pad
true $A[80, 320]$	+0.27 on A_{44}	moves S up	numeric
$[T, \infty)$ oscillatory	not enclosed	the remainder	OPEN
Fourier $1/n$ tail	$HS \sim 0.08$	1400%	KILL route (b)
σ^2 -Plancherel CC^*	$\lambda_{\min} \sim -10$	—	KILL formula

Midpoints match r480: even $\lambda_{\min}(S) = 5.685657 \cdot 10^{-3}$ at $n = 48$, $T = 150$.

6 Outlook at $L = 0.8$

There $P \neq 0$ is the multiplier $\sigma_P(t) = \sum 2\Lambda(n)n^{-1/2} \cos(t \log n)$, not a rank-three update. Π stays rank two. The pair $(A - P, \Pi)$ is still “multiplier plus rank two”; the negative set of $\sigma_A - \sigma_P$ is a union of bands. Whatever closes $S \succ 0$ at $L = 0.3$ must stay explicit under that symbol.

7 Sealed balance

Probe `schur_cert_probe.py`, SPEC_SHA c81a0b2490cfb475. Smoke 14/14 in 0.05 s; full 21/21 in 1.2 s.

8 Kill table

gate	status	note
Fourier route (b)	KILL	$1/n$ tail of C
σ^2 as $\ Au\ _{[-L, L]}$	KILL	not time-limited
IBP envelope as pad	KILL	$0.76 > 0.002$
$n = 16$ FT	already KILL	r480
ε lift	PASS	entire kernel
c_{tail}	PASS	0.2408
interval $S \succ 0$	OPEN	FT + IBP
scramble / anti-list	PASS	$P = 0$; e^{2L} ; no zeros
no RH / not λ_*	PASS	type is REDUCED

9 What this does not do

It does not prove $\lambda_*(0.3) \geq 0$. It does not enclose the Fourier integrals of the interpolants. It does not treat $L > 0.3$. No cofinal quantifier. No RH claim.